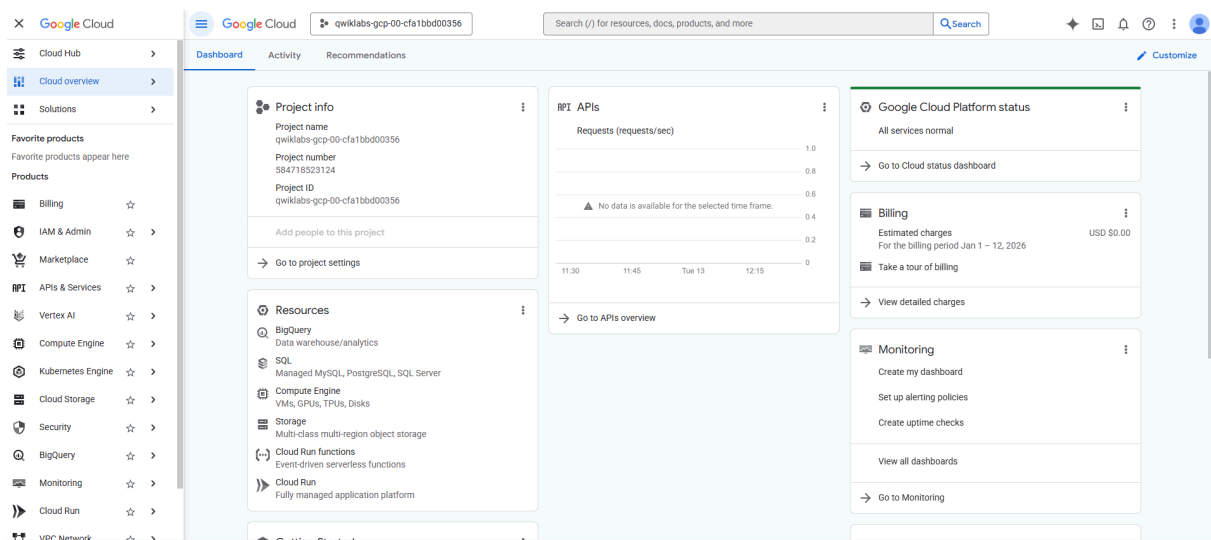


Compute Engine: Qwik Start Windows

Compute Engine lets you create and run virtual machines on Google infrastructure. Compute Engine offers scale, performance, and value that allows you to easily launch large compute clusters on Google's infrastructure.

You can run your Windows applications on Compute Engine and take advantage of many benefits available to virtual machine instances, such as reliable storage options, the speed of the Google network, and Autoscaling.

In this hands-on lab, you learn how to launch a Windows Server instance in Compute Engine and use Remote Desktop Protocol (RDP) to connect to it.

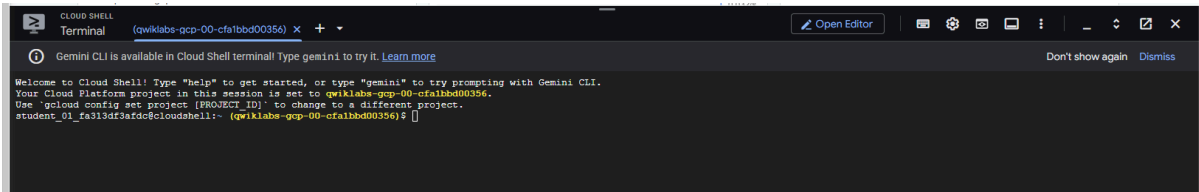
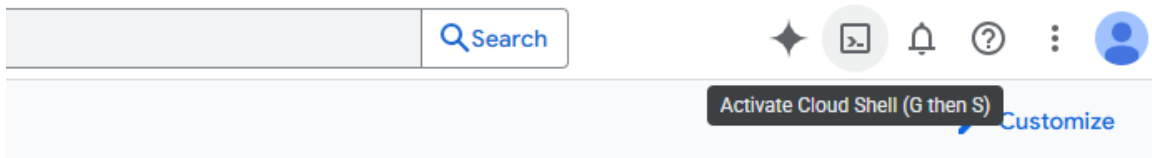


Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

1. Click **Activate Cloud Shell** at the top of the Google Cloud console.





1. Click through the following windows:

- Continue through the Cloud Shell information window.
- Authorize Cloud Shell to use your credentials to make Google Cloud API calls.

When you are connected, you are already authenticated, and the project is set to your **Project_ID**, `PROJECT_ID`. The output contains a line that declares the **Project_ID** for this session:

`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

1. (Optional) You can list the active account name with this command:

```
gcloud auth list
```

Copied!

1. Click **Authorize**.

Output:

```
ACTIVE: *
ACCOUNT: "ACCOUNT"

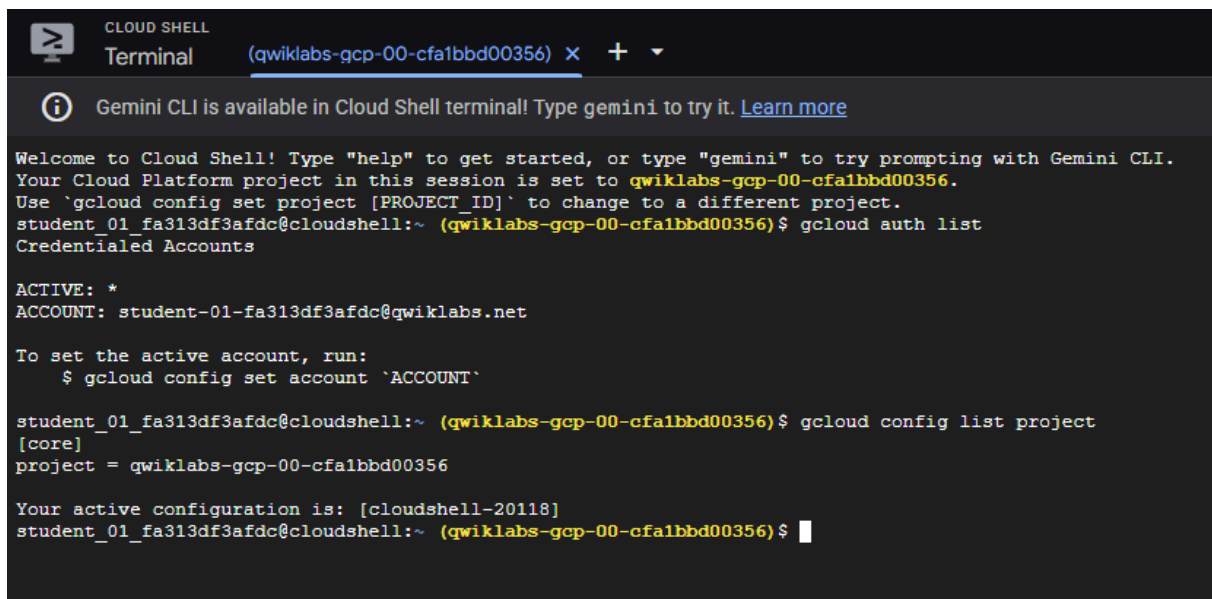
To set the active account, run:
$ gcloud config set account `ACCOUNT`
```

1. (Optional) You can list the project ID with this command:

```
gcloud config list project
```

Output:

```
[core]
project = "PROJECT_ID"
```



The screenshot shows a Cloud Shell terminal window with the title 'CLOUD SHELL Terminal (qwiklabs-gcp-00-cfa1bbd00356)'. A message at the top states: 'Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)'. The terminal text includes a welcome message, the current project ID 'qwiklabs-gcp-00-cfa1bbd00356', and the output of the 'gcloud config list project' command, which shows 'project = qwiklabs-gcp-00-cfa1bbd00356'. It also shows the active account 'student-01-fa313df3afdc@qwiklabs.net' and the active configuration 'cloudshell-20118'.

Create a virtual machine instance

1. In the Cloud Console, on the **Navigation menu** (), click **Compute Engine > VM instances**, and then click **Create Instance**.



2. In the **Machine configuration**.

Select the following values:

Property	Value (type value or select option as specified)
Region	us-west4
Zone	us-west4-a
Series	E2

← Create an instance [Create VM from...](#)

- Machine configuration
e2-medium, us-west4-a
- OS and storage
Debian GNU/Linux 12 (bookworm)
- Data protection
Snapshot schedules
- Networking
1 network interface
- Observability
Install Ops Agent
- Security
- Advanced

Machine configuration

Name *
instance-20260112-233114

Region *
us-west4 (Las Vegas)

Zone *
us-west4-a

Region is permanent Zone is permanent

NEW: Google Axion machine series now in Preview
Try the new N4A series, Google's next generation of Arm-based Axion VMs. [Try now](#)

☒ General purpose ☐ Compute optimized ☐ Memory optimized ☐ Storage optimized ☐ GPUs

Machine types for common workloads, optimized for cost and flexibility

Series	Description	vCPUs	Memory	CPUs
☐ C4	Consistently high performance	2 - 288	4 - 2,232 GB	Intel
☐ C4A	Arm-based consistently high performance	1 - 96	2 - 768 GB	Goo
☐ C4D	Consistently high performance	2 - 384	3 - 3,072 GB	AMC
☐ N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel
☐ N4A	Preview Arm-based flexibility & cost optimization	1 - 64	2 - 512 GB	Goo
☐ N4D	Flexible & cost-optimized	2 - 96	4 - 768 GB	AMC
☐ C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel
☐ C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMC
☒ E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Intel
☐ N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel
☐ N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMC
☐ T2A	Scale-out workloads	1 - 48	4 - 192 GB	Arm
☐ T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMC
☐ N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB	Intel

Monthly estimate

\$28.65
That's about \$0.04 hourly
Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$27.55
10 GB balanced persistent disk	\$1.10
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$28.65

[Compute Engine pricing](#)
[Cloud Operations pricing](#)
[Less](#)

3. Click **OS and storage**.

Click **Change** to begin configuring your boot disk and select the following values:

- **Operating system:** **Windows Server**
- **Version:** **Windows Server 2022 Datacenter**

Click **Select**.

4. Click **Create** to create the instance.

← Create an instance [Create VM from...](#)

- Machine configuration
e2-medium, us-west4-a
- OS and storage
Windows Server 2022 Datacenter
- Data protection
Snapshot schedules
- Networking
1 network interface
- Observability
Install Ops Agent
- Security
- Advanced

Operating system and storage

Name
instance-20260112-233114

Type
New balanced persistent disk

Size
50 GB

Snapshot schedule
default-schedule-1

License type
PAYG (Pay-as-you-go)

Image
Windows Server 2022 Datacenter

If you are using Windows and intend to run additional Microsoft software, please fill out the [License Verification Form](#)

[Learn more](#) about Microsoft license mobility requirements

Change

Additional disks

[+ Add new disk](#) [+ Attach existing disk](#) [+ Add local SSD](#)

Container

Deprecated

Deploy a container image to this VM instance

[Deploy container](#)

Monthly estimate

\$66.63
That's about \$0.09 hourly
Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$27.55
Premium image usage fee*	\$33.58
50 GB balanced persistent disk	\$5.50
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$66.63

* Image usage fee is billed by Google.

[Compute Engine pricing](#)
[Cloud Operations pricing](#)
[Less](#)

Remote Desktop (RDP) into the Windows Server

Test the status of Windows Startup

After a short time, the Windows Server instance will be provisioned and listed on the VM Instances page with a green status icon.



The server instance may not be ready to accept RDP connections, as it takes a while for all OS components to initialize.

1. To see whether the server instance is ready for an RDP connection, run the following command at your Cloud Shell terminal command line and please make to replace `[instance]` with the VM Instance that you created earlier.

```
gcloud compute instances get-serial-port-output [instance] --zone=ZONE
```

1. If prompted, type N and press ENTER.

Repeat the command until you see the following in the command output, which tells you that the OS components have initialized and the Windows Server is ready to accept your RDP connection.

```
-----  
Instance setup finished. instance is ready to use.  
-----
```

```

student_01_fa313df3afdc@cloudshell:~ (qwiklabs-gcp-00-cfa1bbd00356) $ gcloud compute instances get-serial-port-output instance-20260112-233114 --zone=us-west4-a
CSM BBS Table full.
BdsDxe: loading Boot0001 "UEFI Google PersistentDisk " from PciRoot(0x0)/Pci(0x3,0x0)/Scsi(0x1,0x0)
BdsDxe: starting Boot0001 "UEFI Google PersistentDisk " from PciRoot(0x0)/Pci(0x3,0x0)/Scsi(0x1,0x0)

UEFI: Attempting to start image.
Description: UEFI Google PersistentDisk
FilePath: PciRoot(0x0)/Pci(0x3,0x0)/Scsi(0x1,0x0)
OptionNumber: 1.
2026/01/12 23:34:48 GCEGuestAgent: GCE Agent Started (version 20251120.01)
2026-01-12T23:35:56.3010Z GCEGuestAgentManager: [INFO]: Running GCEAgentManager as a Windows service
2026-01-12T23:35:56.3017Z GCEGuestAgentManager: [INFO]: Core plugin config file is present, setting skipCorePlugin to [false]
2026-01-12T23:35:56.3387Z GCEGuestAgentManager: [INFO]: Initializing Google Guest Agent...
CSM BBS Table full.
BdsDxe: loading Boot0003 "Windows Boot Manager" from HD(2,GPT,65AE907A-F029-4524-894E-17511032F0AB,0x8000,0x32000)/EFI\Microsoft\Boot\bootmgfw.efi
BdsDxe: starting Boot0003 "Windows Boot Manager" from HD(2,GPT,65AE907A-F029-4524-894E-17511032F0AB,0x8000,0x32000)/EFI\Microsoft\Boot\bootmgfw.efi

UEFI: Attempting to start image.
Description: Windows Boot Manager
FilePath: HD(2,GPT,65AE907A-F029-4524-894E-17511032F0AB,0x8000,0x32000)/EFI\Microsoft\Boot\bootmgfw.efi
OptionNumber: 3.
2026-01-12T23:36:25.1081Z GCEGuestAgentManager: [INFO]: Running GCEAgentManager as a Windows service
2026-01-12T23:36:25.2253Z GCEGuestAgentManager: [INFO]: Core plugin config file is present, setting skipCorePlugin to [false]
2026-01-12T23:36:25.4504Z GCEGuestAgentManager: [INFO]: Initializing Google Guest Agent...
2026-01-12T23:36:25.5315Z GCEGuestAgentManager: [INFO]: Running Guest Agent setup with config: {Version:20251120.01 EnableACSWatcher:true CorePluginPath:C:\Prog
79467525 svcActPresent:true}
2026-01-12T23:36:25.5315Z GCEGuestAgentManager: [INFO]: Registered ACS watcher and handler
2026-01-12T23:36:26.5900Z GCEGuestAgentManager: [INFO]: Initializing plugin manager for instance "4126864701479467525", agent version: "20251120.01"
2026-01-12T23:36:26.6891Z GCEGuestAgentManager: [INFO]: Plugin manager initialized
2026-01-12T23:36:26.6891Z GCEGuestAgentManager: [INFO]: Google Guest Agent (version: "20251120.01") Initialized...
2026-01-12T23:36:26.7355Z GCEGuestAgentManager: [INFO]: Creating metric registry for "events_manager" with flush interval: 1m0s, max records: 10
2026-01-12T23:36:26.6891Z GCEGuestAgentManager: [INFO]: Transitioning windows service manager to running
2026-01-12T23:36:26.9503Z GCEGuestAgentManager: [INFO]: Instance has service account: true, setting ACS client isEnabled to true
2026-01-12T23:36:26.1827Z GCEGuestAgentManager: [INFO]: Cloud logging initialized (GCEWindowsAgentManager.exe)
2026/01/12 23:36:26 GCEGuestAgent: GCE Agent Started (version 20251120.01)
2026/01/12 23:36:26 GCEGuestAgent: Initializing MDS mTLS bootstrapping handler
2026/01/12 23:36:26 GCEGuestAgent: Starting the scheduler to run jobs
2026/01/12 23:36:26 GCEGuestAgent: Scheduler - start: []
2026/01/12 23:36:26 GCEGuestAgent: Starting the scheduler to run jobs
2026/01/12 23:36:26 GCEGuestAgent: Scheduling job: telemetryJobID
2026/01/12 23:36:26 GCEGuestAgent: Scheduling job "telemetryJobID" to run at 24.000000 hr interval
2026/01/12 23:36:26 GCEGuestAgent: Scheduler - added: [now 2026-01-12 23:36:26.7524826 +0000 GMT entry 1 next 2026-01-13 23:36:26 +0000 GMT]
2026-01-12T23:36:27.3740Z OSConfigAgent Info: OSConfig Agent (version 20251028.00.0+win64) started.
2026-01-12T23:36:27.3740Z OSConfigAgent Info: Writing repo file C:\ProgramData\GoGet\repos/google_osconfig_managed.repo with updated contents
2026-01-12T23:36:27.5107Z OSConfigAgent Info: Beginning ApplyConfigTask.
2026-01-12T23:36:27.6235Z OSConfigAgent Info: Executing policy "goog-ops-agent-policy"
2026-01-12T23:36:27.6547Z OSConfigAgent Info: Validate: resource "add-repo" validation successful.
2026-01-12T23:36:27.6886Z OSConfigAgent Info: Check state: resource "add-repo" state is NON_COMPLIANT.
2026-01-12T23:36:27.7255Z OSConfigAgent Info: Enforcing repo C:\ProgramData\GoGet\repos\osconfig_managed_48053a8de9.repo.
2026-01-12T23:36:27.7831Z OSConfigAgent Info: Enforce state: resource "add-repo" enforcement successful.
2026-01-12T23:36:27.8476Z OSConfigAgent Info: Validate: resource "install-pkg" validation successful.
2026-01-12T23:36:28.3880Z OSConfigAgent Info: Check state: resource "install-pkg" state is NON_COMPLIANT.
2026-01-12T23:36:28.4193Z OSConfigAgent Info: Installing goget package "google-cloud-ops-agent"
2026/01/12 23:36:28 GCEInstanceSetup: WinRM certificate details: Subject: CN=instance-20260112-233114, Thumbprint: 1C8BD8BC5C57A74EE6AE143E48DC5410B2249C9D
2026/01/12 23:36:29 GCEInstanceSetup: RDP certificate details: Subject: CN=instance-20260112-233114, Thumbprint:
2026/01/12 23:36:30 GCEInstanceSetup: Microsoft Windows PAYG license 4079807029871201927 found.
2026/01/12 23:36:30 GCEInstanceSetup: Key Management Service machine name set to kms.windows.googlecloud.com successfully.
2026/01/12 23:36:30 GCEInstanceSetup: Microsoft Windows PAYG license 4079807029871201927 found.
2026/01/12 23:36:30 GCEInstanceSetup: Key Management Service machine name set to kms.windows.googlecloud.com successfully.
2026/01/12 23:36:34 GCEInstanceSetup: Installed product key WX4NM-KWYW-QJNR4-IVQCB-6V63 successfully.
2026/01/12 23:36:34 GCEInstanceSetup: Activating instance...
2026/01/12 23:37:00 GCEInstanceSetup: Activating Windows(R), ServerDatacenter edition (ef6cf9f-8c5d-44ac-9aad-d6a2a0ae03) ...
2026/01/12 23:37:01 GCEInstanceSetup: Product activated successfully.
2026/01/12 23:37:02 GCEInstanceSetup: Activation successful.
2026/01/12 23:37:02 GCEInstanceSetup: Running 'schtasks' with arguments '/change /tn GCSEStartup /enable'
2026/01/12 23:37:03 GCEInstanceSetup: --> SUCCESS: The parameters of scheduled task "GCSEStartup" have been changed.
2026/01/12 23:37:03 GCEInstanceSetup: Running 'schtasks' with arguments '/run /tn GCSEStartup'
2026/01/12 23:37:03 GCEInstanceSetup: --> SUCCESS: Attempted to run the scheduled task "GCSEStartup".
2026/01/12 23:37:03 GCEInstanceSetup: Instance setup finished. instance-20260112-233114 is ready to use.
2026-01-12T23:37:04.4012Z [INFO]: Initialized Metadata Script Runner Compat (version 20251120.01)
2026-01-12T23:37:04.4642Z [INFO]: Enable core plugin set to: [false], launching script runner for event "startup" from "C:\Program Files\Google\Compute Engine\metadata_scripts\GCEMetadataScripts.exe"
2026/01/12 23:37:06 C:\Program Files\Google\Compute Engine\metadata_scripts\GCEMetadataScripts.exe: Starting startup scripts (version 20251009.01).
2026-01-12T23:37:06.4096Z [INFO]: 2026/01/12 23:37:06 C:\Program Files\Google\Compute Engine\metadata_scripts\GCEMetadataScripts.exe: Starting startup scripts (version 20251009.01).
2026/01/12 23:37:06 C:\Program Files\Google\Compute Engine\metadata_scripts\GCEMetadataScripts.exe: No startup scripts to run.
2026-01-12T23:37:06.5372Z [INFO]: 2026/01/12 23:37:06 C:\Program Files\Google\Compute Engine\metadata_scripts\GCEMetadataScripts.exe: No startup scripts to run.
2026-01-12T23:37:26.8826Z OSConfigAgent Info: Enforce state: resource "install-pkg" enforcement successful.
2026-01-12T23:37:27.0118Z OSConfigAgent Info: Check state post enforcement: resource "add-repo" state is COMPLIANT.
2026-01-12T23:37:27.1223Z OSConfigAgent Info: Policy "goog-ops-agent-policy" resource "add-repo" state: COMPLIANT
2026-01-12T23:37:27.3508Z OSConfigAgent Info: Check state post enforcement: resource "install-pkg" state is COMPLIANT.
2026-01-12T23:37:27.4412Z OSConfigAgent Info: Policy "goog-ops-agent-policy" resource "install-pkg" state: COMPLIANT
2026-01-12T23:37:27.5630Z OSConfigAgent Info: Successfully completed ApplyConfigTask

```

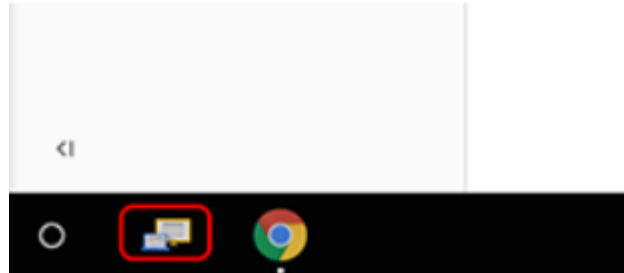
RDP into the Windows Server

1. To set a password for logging into the RDP, run the following command in Cloud Shell. Be sure you replace `[instance]` with the VM Instance that you created and set `[username]` as `admin`.

```
gcloud compute reset-windows-password [instance] --zone ZONE --user [username]
```

1. If asked `Would you like to set or reset the password for [admin] (Y/n)?`, enter Y. Record the password for use in later steps to connect.

2. Connect to your server. There are different ways to connect to your server through RDP, depending on whether you are on Windows or not:
 - If you are using a Chromebook or other machine at a Google Cloud event there is likely an RDP app already installed on the computer. Click the icon as below, if it is present, in the lower left corner of the screen and enter the external IP of your VM.



1.
 - If you are not on Windows but using Chrome, you can connect to your server through RDP directly from the browser using the [Spark View](#) extension. Click on **Add to Chrome**. Then, click **Launch app**.
2. Once launched, the **Spark View (RDP)** window opens. Use your Windows username **admin** and password you previously recorded in Step 2.
3. Add your VM instance's External IP as your Domain. Click Connect to confirm you want to connect.

If you are on a Macintosh, there are several freely accessible RDP Client packages available to install, such as [CoRD](#). After installing, connect as above to the External IP address of the Windows server. Once it has connected, it will open up a login page where you can specify Windows username **admin** and password from the output of above mentioned command to log in (ignore the Domain: field).

Once logged in, you should see the Windows desktop!

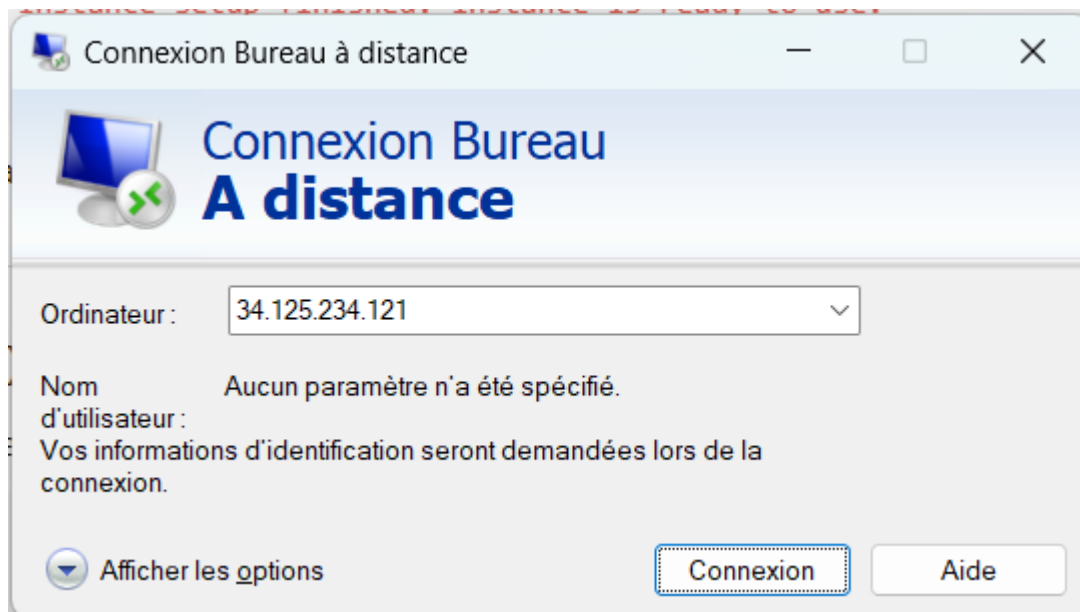
```
student_01_fa313df3afdc@cloudshell:~ (qwiklabs-gcp-00-cfalbbd00356) $ gcloud compute reset-windows-password instance-20260112-233114 --zone=us-west4-a --user admin

This command creates an account and sets an initial password for the
user [admin] if the account does not already exist.
If the account already exists, resetting the password can cause the
LOSS OF ENCRYPTED DATA secured with the current password, including
files and stored passwords.

For more information, see:
https://cloud.google.com/compute/docs/operating-systems/windows#reset

Would you like to set or reset the password for [admin] (Y/n)? Y

Resetting and retrieving password for [admin] on [instance-20260112-233114]
Updated [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-00-cfalbbd00356/zones/us-west4-a/instances/instance-20260112-233114].
ip_address: 34.125.234.121
password: y24--@D_kIq7v
username: admin
```

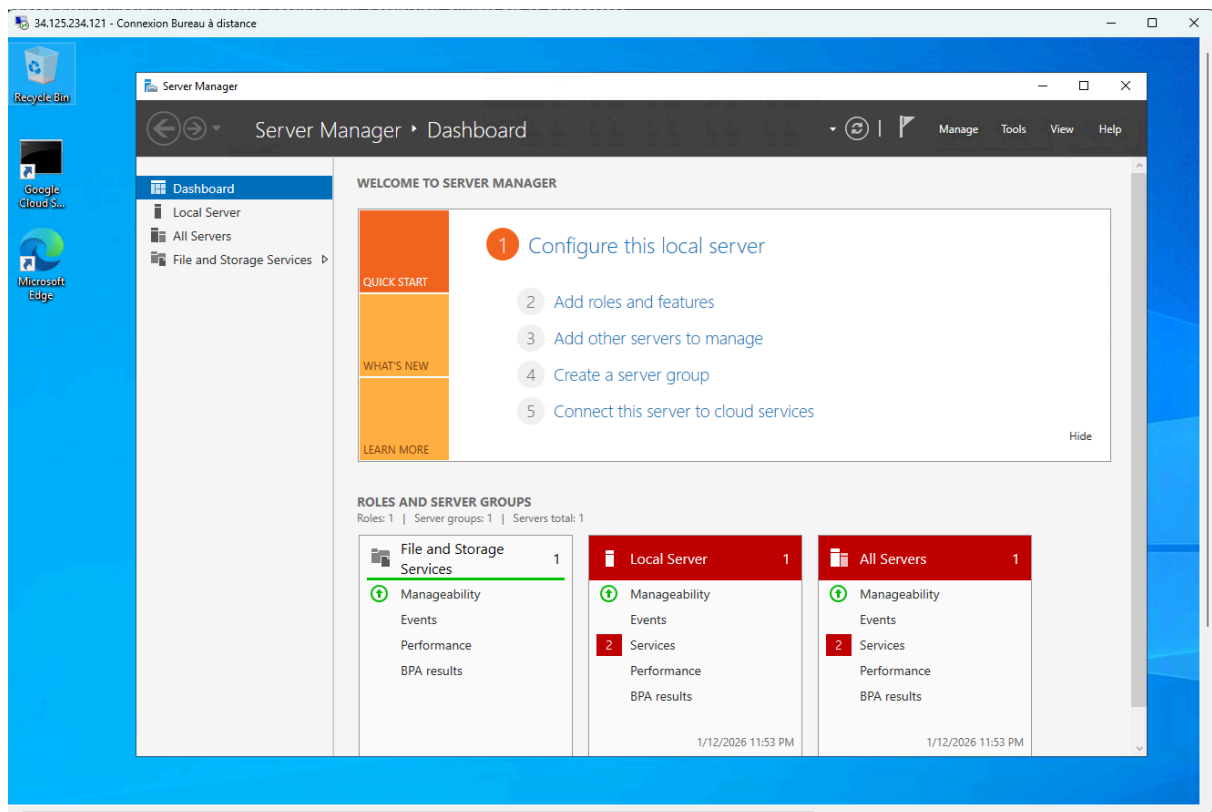


Copy and paste with the RDP client

Once you are securely logged into your instance, you may find yourself copying and pasting commands from the lab manual.

To paste, hold the **CTRL-V** keys (if you are a Mac user, using **CMND-V** will not work.) If you are in a Powershell window, be sure that you have clicked into the window or else the paste shortcut won't work.

If you are pasting into putty, **right click**.



Une VM est comme un ordinateur classique jusque que celui-ci vie dans un datacenter de chez google et on y accede via le reseau

Dans ce lab en selectionnant windows server 2022 data center comme OS cela fait que notre VM va se comporter comme un server physique qui va run programme

une fois que nous avons installer la VM est que celle-ci est ready to use comment bien verifier cela on utiliser la commande et cela va nous renvoyer plein d'information technique et une fois que nous observons le message comme quoi l'iinstance est ready cela est good

```
gcloud compute instances get-serial-port-output [instance] --zone=ZONE
```

Nous avons besoin de nous connecter a notre interface donc nous allons creer une mdp pour notre user admin en utilisant la commande , une fois que nous avons le mdp et le user nous allons utiliser le protocole (RDP Remote Desktop Protocol) qui est le bureau distant et cela va ouvrir Windows Server